Protocol Development and Quality Improvement

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Learning Objectives

1. Determine the steps involved in the development of a guideline and/or policy.
2. Design a medication use evaluation (MUE) with selection of an appropriate medication and/or medication-related process suitable for an MUE.
3. Determine the appropriate quality improvement tool to use in order to optimize outcomes in a critically ill patient population.
4. Identify processes or quality improvement initiatives that would benefit from a gap analysis.
5. Describe the types of pharmacotherapeutic interventions and documentation processes to justify the value of clinical pharmacy services.

Abbreviations in This Chapter

ADE	Adverse drug event
ASHP	American Society of Health-System Pharmacists
CMS	Centers for Medicare & Medicaid Services
CPOE	Computerized provider order entry
CPS	Clinical pharmacy services
DUE	Drug use evaluation
EBM	Evidence-based medicine
EHR	Electronic health record
ICU	Intensive care unit
LOS	Length of stay
MUE	Medication use evaluation
PAI	Practice Advancement Initiative
PI	Performance improvement
P&T	Pharmacy and therapeutics (committee)
QA	Quality assurance
QI	Quality improvement
SUP	Stress ulcer prophylaxis
TJC	The Joint Commission

Self-Assessment Questions

Answers and explanations to these questions may be found at the end of this chapter.

1. As the clinical coordinator/manager, you have received a request to add the critical care pharmacist's review of all intensive care unit (ICU) patients. You recognize that you lack the necessary personnel to accomplish this request. Which justification method would offer the most significant financial impact on the institution to justify the position?
   A. Developing protocols in the critically ill patient.
   B. Reducing prescribing errors.
   C. Reducing medication administration errors.
   D. Implementation of an anticoagulation reversal stewardship program.

2. You are the critical care pharmacist implementing a delirium screening tool in critically ill patients according to the Society of Critical Care Medicine guidelines. Which would best be described as a practice environment barrier to implementing delirium screening in your ICU?
   A. The nursing standard work does not include delirium screening.
   B. Screening process takes up valuable nursing time.
   C. Providers are unfamiliar with how to perform delirium screening.
   D. Providers believe their knowledge of bedside delirium is adequate and does not require a screening tool.

3. The pharmacy and therapeutics (P&T) committee would like to evaluate the use of pharmacotherapy in stress ulcer prophylaxis (SUP) in the ICU. Which is the best method for making this evaluation?
   A. Perform a medication use evaluation (MUE).
   B. Administer a performance improvement (PI) initiative.
   C. Review adverse drug event (ADE) data from the ICU.
   D. Review medication error data from the ICU.

4. Which medication use process is best suited for an MUE?
   A. Pharmacist verification times for routine orders in the ICU.
   B. Management of warfarin-induced hypoprothrombinemia.
   C. Drug interaction warnings on the computerized provider order entry (CPOE) system.
   D. Duplicate warnings on the CPOE system.

5. You have been asked to develop a quality improvement (QI) strategy for your ICU regarding when to institute targeted temperature management in patients after cardiac arrest. The current process has been defined, and best practices for targeted temperature management have been outlined. You identify key stakeholders in this patient population to collaborate with you in this process improvement. Which is the next best step in developing the QI program?
   A. Determine which patients with cardiac arrest should receive targeted temperature management.
   B. Evaluate the current clinical guidelines associated with targeted temperature management.
   C. Determine the current process of targeted temperature management.
   D. Evaluate the data associated with patients receiving targeted temperature management.

6. You obtain data related to time to administration of immediate antibiotics for sepsis from your CPOE system in order to determine a baseline metric for quality improvement. Using the define, measure, analyze, improve, control (DMAIC) method, which step describes the process phase to determine the baseline time to antibiotics?
   A. Data are collected.
   B. Methods are established.
   C. Change process is implemented.
   D. Data are analyzed.

7. Which metric best quantifies a critical care pharmacist's activities?
   A. Weighing each variable to quantify measured activities.
   B. Number of cardiac arrest codes attended each month.
   C. Number and severity of hospital-acquired *Clostridioides difficile* infections each year.
   D. Number of in-services provided each year.

8. As the clinical pharmacist in the ICU, you are asked to evaluate the quality initiatives specific to critically ill patients. Which best describes a Centers for Medicare & Medicaid Services (CMS) quality metric?
   A. Sepsis management bundle and venous thromboembolism prophylaxis.
   B. Thrombolytic therapy for stroke and SUP.
   C. Heart failure readmission and pneumonia vaccination rates.
   D. Ventilator bundles and acute coronary syndrome readmission rates.

BPS Critical Care Pharmacy Examination Content Outline

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 3: Professional Practice
    a. Quality of Care
        i. 3A2, 3A3, 3A4, 3A7
    b. Practice Management
        i. 3B1

I. POLICY AND GUIDELINE DEVELOPMENT

Objective: Critical care pharmacists should participate in developing ICU and institutional policies, procedures, guidelines, and education.

A. Policy and Procedure
   1. Policy – A course or plan of action; written policies establish standards of practice or quality/compliance measures and protects against error. Policies intend to show the "why" behind an action and include a description of "what," "when," "who," or "where" actions should occur in a specific situation. Most policies are mandatory and may result in disciplinary action if not followed. Policy is based on well-defined standards including those that may be required to comply with laws, regulations, or contracts.
      a. Benefits
         i. Improves consistency and efficiency of work processes
         ii. Provides a training tool during a new employee's orientation
         iii. Provides reference material for education and practice
         iv. Minimizes practice variations
         v. Reduces the organizational risk by mandating compliance
      b. Development and implementation should involve all relevant team members to ensure all concerns are addressed and to promote input from all stakeholders.
      c. Examples of policies:
         i. Acute pulmonary embolism response team
         ii. High-alert medication list and safety standards
         iii. Pharmacist initiated automatic therapeutic interchange
         iv. Pharmacist-initiated intravenous to oral conversion
   2. Procedure – A series of actions or tasks performed in a specific way or sequence that is intended to achieve a desired result or accomplish a task in a consistent manner; the "how" to achieve the desired outcome or comply with a stated policy.
      a. Requires approval by the organizational committee
      b. Describes each team member's responsibility
      c. Defines the expected outcome(s)
      d. Procedures may be described in the following forms:
         i. Written instructions
         ii. Flowcharts
         iii. Checklists
      e. Can be used as a QI tool or a source of measures
      f. Examples of procedures:
         i. Central line placement
         ii. Medication event reporting
         iii. Use of the AddEase Administration System
         iv. Downtime procedures during system outages or natural disasters

B. Clinical Guidelines/Protocol – Evidence-based statements that assist practitioners with decisions for specific clinical circumstances. Should include all aspects of care including the following: Patients that are excluded from the treatment or have contraindications, initiation, dosing (if applicable), titration, monitoring, education, duration of therapy, etc.
   1. Supports clinical decision-making by defining best practice
   2. Uses evidence-based and standardized treatment options
   3. Developed by examining the evidence and gaining consensus among practitioners

4. Interdisciplinary team effort: Patient, physician/prescriber, nurse, respiratory therapist, pharmacist, clinical dietitian, physical therapist, social worker and/or case manager, hospital administrators, and patient's family/support network
5. Voluntary – Allows for deviation depending on clinical scenario
6. Protocol can be an extension of a clinical policy or procedure.
7. Clinical guidelines and protocols are often kept separate from institutional policies and procedures.
8. Examples of disease management guidelines include:
    a. Pulmonary embolism therapeutic management
    b. Pain, Agitation, Delirium, Immobility, and Sleep (PADIS) management
    c. Spontaneous awakening and breathing trials
    d. Anticoagulation reversal
9. Critical care pathways
10. Formulary proposals
11. Medication reconciliation for admission and transfer within the health care system
12. Clinical decision support
    a. Consider ways to incorporate guidelines/protocols into CPOE to facilitate ordering and monitoring in conjunction with the guideline/protocol.

C. Framework of a Policy and/or Procedure

Critical care pharmacists should lead, or be actively involved with, the assessment of medication-related guidelines, protocols, practice changes, or PI initiatives in the ICU.

1. Purpose – For whom the policy is intended, why the policy is needed, and the desired goals
2. Definitions – Provide the reader with a clear understanding of the language
3. Intent of the policy
4. Provisions
5. Procedure
6. Resources/references
7. Owner
8. Date created/modified/last reviewed
9. Approvals

D. How to Develop Policies

1. Planning
    a. Define and prioritize policy goals and objectives.
    b. Identify key stakeholders and include them in the policy development process.
    c. Appoint a project facilitator/leader.
    d. Clearly delineate each team member's roles and responsibilities in the policy development process.
    e. Develop a strategy for tracking and managing each group's progress.
2. Consultative process for developing a particular policy
    a. Solicits input from organizational or external stakeholders
    b. Promotes good relations between sites and service providers
    c. Encourages participation and feeling of ownership in the process
    d. Provides new ideas and expertise
    e. Alternative viewpoint may identify inconsistency, ambiguity, and/or duplication.
3. General steps in drafting
    a. Write a first draft.
    b. Distribute the draft, and consult across the same organization/stakeholders/experts for comments.
    c. Review feedback, both written and verbal, and amend/revise the draft.
    d. Redistribute the draft for additional/final feedback.

e. Write the final draft.
f. Document the date that the policy was ratified and then subsequent review dates.
g. Committee review and approval.
h. Incorporate the new policy into the policy and procedures.
i. Communicate the new policy to all relevant people.

4. Identify policy review process.
   a. Formal process may be required within the institution (i.e., standing policy and procedures committee, which develops and reviews the policy and provides recommendations and decisions).
   b. Involvement and approval from the P&T and critical care committees or other oversight committees

E. Education
1. Create education for the key stakeholders (e.g., physicians, advanced practice providers, nurses, pharmacists, dietitian, respiratory therapists).
2. Disseminate education through newsletters, intranet posts, in-services, huddles, quarterly updates, grand rounds, one-on-one direct education, online learning modules, etc.
3. Assessment of education through annual competency examination, skills assessment, monthly simulation modules, etc.

F. Review of existing policies. Continuous QI requires documentation and training for pharmacists and staff. Many state boards of pharmacy are requiring reportable events to be identified and documented. Policies and procedures can also be developed for analyzing the data collected to assess causes and contributing factors so that findings can be used to improve outcomes.
1. Every policy should regularly be reviewed for relevance and appropriateness (e.g., every 1–3 years), depending on organizational structure/standards.
2. Monitor and evaluate compliance with, and effect on, policies and guidelines.
3. Evaluate the policy on a routine schedule
   a. Determine need for updates to the policy.
   b. Evaluation strategies
   c. Ongoing monitoring
   d. Presentation of data
   e. Consumer feedback
   f. Stakeholder feedback (e.g., physician, pharmacy, nursing, policy steering committee, critical care committee)
   g. Planning day agenda
   h. Forms: Data collection sheets, data reporting format, questionnaires
   i. Key stakeholder questionnaires
4. Evidence-based critical care literature and clinical practice guidelines in designing a patient-specific plan of care
   a. Definition: "The conscientious, explicit and judicious use of current best evidence in making decisions about the care of individual patients. The practice of evidence-based medicine means integrating individual clinical expertise with the best available external clinical evidence from systematic research." (BMJ 1996;312:71)
   b. Pros and cons of evidence-based medicine (EBM)
      i. Advantages
         (a) Critical appraisal skills of the literature improve with the review of EBM.
         (b) Wasteful, outdated, or harmful practices can be abandoned.
         (c) Presupposes that we keep up-to-date; ideally, a systematic process of incorporating new EBM is included

(d) Makes the decision-making process transparent to colleagues and patients
(e) Leads to greater appreciation of the evidence for our practice as well as the inherent uncertainties

ii. Disadvantages
(a) Time-consuming
(b) Sometimes impossible (when there is no published literature on a question)
(c) Useful papers may be disregarded because of minor blemishes (rescue bias).
(d) Contradictory study findings leading to differences in interpretation of benefit
(e) No science to tell us how robust the evidence must be for it to be incorporated into clinical practice
(f) External validity is subjective, and evidence can be misapplied.
(g) Easy-to-prove techniques more favored in literature
(h) It is never "up-to-date."
(i) Tends to emphasize the priority of randomized controlled trials (which have inherent flaws) to the exclusion of other study designs (which may be appropriate in certain settings)
(j) May underemphasize patient values and interests
(k) Publications with favorable results are more likely to be published than are those without favorable results (publication bias).

5. Barriers to implementing evidence
a. Practice environment (organizational context)
i. Financial disincentives – Lack of reimbursement
ii. Organizational constraints – Lack of time; insufficient leadership or administrative support; staffing, equipment, or resource constraints
iii. Perception of liability – Risk of formal complaint
iv. The patient's expectations – Expressed wishes related to care
b. Prevailing opinion (social context)
i. Standard work – Usual routine
ii. Opinion leaders – Key individuals not in agreement with evidence
iii. Medical training – Obsolete knowledge
iv. Advocacy – By pharmaceutical companies
c. Knowledge, skills, and attitudes (professional context)
i. Clinical uncertainty – Necessary test for vague symptoms
ii. Sense of competence – Self-confidence in skills
iii. Compulsion to act – Need to do something
iv. Information overload – Inability to appraise evidence

G. Checklist to Follow When Developing Guidelines
1. Scope and purpose
a. Objectives are specifically described.
b. Health questions covered by the guideline are specifically described.
c. The patient population to which the guideline applies is specifically described.
2. Stakeholder involvement
a. Guideline development group includes individuals relevant to the guideline.
b. Target population's views and preferences have been identified.
c. Target users of the guideline are defined.
3. Development of guideline
a. Literature search is systematic.
b. Criteria for selecting evidence are clearly defined.

c. Strengths and limitations of evidence are described.
d. Health benefits, adverse effects, and risks are considered when developing recommendations.
e. Link between recommendations and supporting evidence is provided.
f. Guideline is externally reviewed by experts.
g. Defined times for guideline updates are provided.

4. Clarity of recommendations
   a. Recommendations are clear and unambiguous.
   b. Different options are clearly presented.
   c. If involving medication therapy, dosing is clear, including initial starting rate and titrations for continuous infusions.
5. Applicability
   a. Guideline describes facilitators and barriers to the application.
   b. Guideline provides tools on how it should be applied to practice.
   c. Guideline presents monitoring and/or auditing criteria.
6. Approval
   a. Guidelines involving the use of medications should obtain appropriate committee-level approvals prior to implementation.
7. Monitoring of adherence to the guidelines through MUEs to assess compliance or need for revisions.

H. Other considerations
   1. All policies, procedures, guidelines/protocols should be accessible during periods of electronic health system maintenance and downtime (planned and unplanned).
   2. Important to ensure each unit where these documents are used or at the institutional or health-system level are kept in a location where they can be easily accessed should a mass casualty event or downtime occur

Patient Case

1. As the critical care pharmacist, you have been asked to develop a guideline for reversal agents used in the setting of bleeding associated with anticoagulation therapy. Which is the best example of a component that should be involved in establishing the clinical pathway?
   A. Evaluate the closed-loop technology to support a clinical pathway, and establish a physician champion.
   B. Use EBM to support the clinical pathway and develop agreement among practitioners.
   C. Develop a clinical protocol, and obtain agreement among practitioners.
   D. Evaluate formulary proposals and the EBM supporting the clinical practice.

II. GAP ANALYSIS

A. A gap analysis is an assessment of a practice model that may be within your health system or pharmacy and that is compared with a best practice model (actual vs. potential performance). Within pharmacy, a gap analysis may focus on pharmacy services, pharmacy technology, or a specific medication or medication process. A gap analysis can also be used to analyze gaps in processes and between the existing outcome and the desired outcome. This process can be summarized as follows.

B. Goals of a Gap Analysis – To provide the project team with an understanding of the differences between current and best practices. An assessment must be made of the barriers that exist before best practice can be implemented successfully.
   1. Review systems.
   2. Develop requirements.
   3. Comparisons
   4. Implications
   5. Recommendations

C. Gap Analysis
   1. Identify the existing process or gap analysis questions.
   2. Identify the existing outcome.
   3. Identify the desired outcome.
   4. Identify the process for achieving the desired outcome.
   5. Assess the response to the level of compliance or implementation (e.g., fully, partly, in progress, no activity, or noncompliant).
   6. Develop an action plan to fill the gap.
   7. Develop and prioritize the requirements, and develop a timeline for completion to bridge the gap.

D. Best Practice Evaluation
   1. Review of the primary literature
   2. Survey practitioners – Feedback from practitioners (e.g., ICU pharmacists, nurses, physicians)
   3. Benchmarking against gold standards such as:
      a. ISMP Best Practices
      b. ISMP Safety Alert newsletters
      c. Published guidelines such as USP 797 or USP 800
      d. The Joint Commission and National Patient Safety Goals
      e. CMS Core Measures
   4. Survey strategies
      a. Have a single well-defined objective.
      b. Keep the survey short.
      c. Assure respondents that survey responses will be kept anonymous (if possible).
      d. Survey a large-enough sample to ensure meaningful results.
      e. Explain the purpose of the survey, and how you intend to use the results, in your invitation/greeting page.
      f. Spell out the time expectations in your invitation and on the greeting page.
      g. Survey the appropriate people.
      h. Design the survey for easily measurable results.
      i. Evaluate one concept per question.
      j. Avoid biasing the response.
      k. Limit the number of required questions.
      l. Question order matters – The first question or two should be easy and interesting.
      m. Create a logical flow to the questions.
      n. Test the survey.
      o. Share the results and actions with the respondents.

E. A gap analysis for medication safety should include strategies for prevention and mitigation, assessment and detection, therapeutic use, critical thinking and knowledge, and education. A gap analysis should also include policies, procedures, protocols, guidelines, competencies, patient care models, educational methods, and other key components to maximize efficacy and prevent medication errors and harm.

F. Examples of a Gap Analysis Include:
   1. Centers for Disease Control and Prevention (CDC): The Core Elements of Hospital Antibiotic Stewardship Programs: Antibiotic Stewardship Program Assessment Tool (excerpt) (Appendix 1)
   2. Opioid ADE prevention
   3. Anticoagulation agent ADE prevention
   4. Aminoglycoside ADE prevention
   5. Intravenous-to-oral conversion of antihypertensives
   6. Medication labeling
   7. Medication barcoding and scanning
   8. Narcotic diversion prevention
   9. Sterile admixture services
   10. Extemporaneous compounding services
   11. Pharmacist activities during a code or rapid response
   12. Pharmacy patient care models – Evaluating the number of ICU pharmacists to patient ratios based on workload, documentation requirements
   13. American Society of Health-System Pharmacists (ASHP) Practice Advancement Initiative (PAI) to optimize care through pharmacist-provided comprehensive medication management
   14. ICU liberation practices, including ABCDEF bundle
   15. Antibiotic streamlining
   16. Criteria-based antimicrobials and non-antimicrobials for high-cost/misused drugs
   17. Departmental professional development education
   18. Institute for Safe Medication Practices (ISMP) Targeted Medication Safety Best Practices for Hospitals
      a. Can be used to assess progress on safety recommendations from ISMP
      b. Can also be used to develop metrics related to implementation status of these best practices.
   19. The Joint Commission (TJC) Medication Management Standards

Patient Case

2. Neuromuscular blocking agents are high alert medications and have led to significant medication errors and patient harm. You noticed at your new job in the medical ICU when you went to pull rocuronium from the automated dispensing cabinet that there were no warnings on the pyxis, and it was in an open cubie pocket. You recalled ISMP having safety alerts and even a best practice regarding the safe storage and handling of neuromuscular blocking agents. Which is the best example of how to conduct a gap analysis?
   A. Conduct a survey of other institutions to see how they are handling the storage of NMBs and compare with practice at your new institution.
   B. Recommend implementation of storage solutions based on what your previous institution did.
   C. Create a table of the ISMP best practice recommendations for the safe storage and handling of NMBs and compare with the current practice at your new institution.
   D. Survey the staff as to what they think the safest storage would be while still being convenient to obtain.

III. QUALITY ASSURANCE, QUALITY/PERFORMANCE IMPROVEMENT

A. Overview
1. Quality improvement (QI) consists of systematic and continuous actions that lead to measurable improvement in health care services and the health status of targeted patient groups.
2. An organization's quality is based on the current system (e.g., how things are currently done).
3. Health care performance is defined by an organization's efficiency, outcome of care, and level of patient satisfaction. Using benchmarks may help with measurements/goals/outcomes.
4. To achieve a different level of performance (i.e., results) and improve quality, an organization's current system needs to be challenged/evaluated.
5. Key components of a successful QI program:
   a. QI works as systems and processes.
   b. Keeps the intended outcome as the central focus
   c. Embraces a team approach
   d. Uses data to guide decision-making
   e. Focuses on processes versus specific people's actions
6. Improvement strategies
   a. Understand the delivery system and key processes.
   b. Recognize that resources (inputs) and activities carried out (processes), including the work around for the actual defined processes, are addressed together to ensure or improve the quality of care (outputs/outcomes).
7. Quality management departments within a health care institution often share data with the risk assessment department.
8. QI programs within an institution
   a. Executive steering committee
   b. Various departments involved in QI initiatives:
      i. Pharmacy department
      ii. Nursing
      iii. Medical staff
      iv. Respiratory
      v. Quality department
      vi. Medical ethics committee
      vii. P&T committee
      viii. Data reporting
      ix. Medication safety committee
      x. Compliance department
      xi. Critical care/clinical specialty committee
9. Analyzing the quality assurance (QA)/QI program
   a. A normal level (upper and lower control limit) should be established for a process to operate.
   b. The process is evaluated, and the results are compared with the normal level expected. Control charts can show the variance of the output of a process over time. The process is considered *in control* if the variance between measurements is the normal random variation considered inherent in the process. If the variance falls outside the limits or has a run of non-natural points, the process is considered *out of control*.
   c. Example: Established process for daily sedative interruption for mechanically ventilated patients. The preestablished expected level of daily sedative interruption was established when the protocol was considered initiated. The QA data collected evaluate the frequency by which the assessments are being performed over a time interval.

B. Critical Care Pharmacist's Role in QI – Metrics for evaluating the quality of critical care pharmacy services
   1. ICU LOS
   2. Hospital LOS
   3. ICU readmissions
   4. Impact on mortality or morbidity
   5. Impact on disease identification, disease prevalence, or clinical outcome
   6. Evaluation of infectious diseases within the ICU
   7. Cost of care versus diagnosis-related group (DRG) reimbursement
   8. Duration of mechanical ventilation
   9. Medication management procedures need to be continuously monitored and improved because of their complexity.
   10. Medication safety – Adverse drug events prevented, time since event reached a patient, etc.
   11. Direct costs (medication costs, technological upgrades, and software)
   12. Indirect costs (salaries, power for building)
   13. Data collection, analysis
   14. Implement processes to reduce medication costs and waste.

C. National Quality Initiatives
   1. The Institute of Medicine – Chartered in 1970. Published reports titled "The Urgent Need to Improve Health Care Quality," "Crossing the Quality Chasm," "To Err Is Human"
   2. The Institute for Healthcare Improvement (IHI) – Founded in 1991
      a. 100,000 Lives Campaign, 5 Million Lives Campaign (December 12, 2006 – December 9, 2008)
      b. No needless list
         i. No needless deaths
         ii. No needless pain or suffering
         iii. No helplessness in those served or serving
         iv. No unwanted waiting
         v. No waste
         vi. No one left out
      c. Critical care initiatives
         i. Acute myocardial infarction
         ii. Catheter-associated urinary tract infections
         iii. Central line–associated bloodstream infections
         iv. Health care–associated infections
         v. Sepsis detection and initial management
         vi. Medication reconciliation to prevent ADEs
         vii. Pain, agitation/sedation, delirium, immobility, and sleep (PADIS) (i.e., the ABCDEF bundle)
         viii. Surgical site infections
         ix. Ventilator-associated pneumonia
         x. Rapid response teams
         xi. High-alert medication safety
         xii. Pressure ulcer care
      d. The five rights of medication administration - the right patient, the right drug, the right dose, the right route, and the right time
         i. Has now been updated to the seven rights: right person, right medication, right time, right route, right reason, right documentation.
      e. Ask "Why" five times to get to the root cause - key to solving a problem

3. National Quality Forum – Created in 1999
    a. More than 300 measures, indicators, events, practices, and other products to help assess quality. Has been endorsed to become the gold standard of measuring health care quality
    b. Patient safety–endorsed measures notable for ICU care (2015)
        i. ADEs
        ii. Catheter-associated urinary tract infections
        iii. Central line–associated bloodstream infections
        iv. Surgical site infections
        v. Ventilator-associated pneumonia
        vi. Ventilator-associated events
        vii. Stroke
        viii. Acute myocardial infarction
        ix. Falls
        x. Pressure ulcer rate
4. The Leapfrog Group – Launched in 2000
    a. Hospital quality and safety survey – Voluntary survey of hospitals rating themselves on quality and safety practices
    b. Reported at www.leapfroggroup.org
    c. ICU-related measures, health care–associated infections
        i. Catheter-associated urinary tract infections
        ii. Catheter-associated bloodstream infections
        iii. Inpatient hospital-onset MRSA infections
        iv. Inpatient hospital-onset *C. difficile* infections
        v. Surgical site infections
        vi. Specially trained physicians care for ICU patients (this is from Leapfrog report).
5. The Joint Commission (TJC) – Founded in 1951
    a. One of three CMS accreditation organizations, TJC is an independent, not-for-profit organization that sets the standards for accreditation in health care
    b. Tracer methodology – Method of evaluation done during an on-site survey that traces the health care experiences of a patient while in the hospital
    c. Identifies, tests, and specifies standardization of performance measures
    d. 2022 National Patient Safety Goals - effective January 2022
        i. Identify patients correctly.
            (a) Use at least two ways to identify patients. For example, use the patient's name and date of birth. This is done to make sure that each patient receives the correct medication and treatment.
        ii. Improve staff communication: Send important test results to the right staff person on time.
        iii. Use medications safely.
            (a) Before a procedure, label medications that are not labeled (e.g., medications in syringes, cups, and basins). Do this where medications and supplies are set up.
                (1) Unless an authorized staff member prepares a medication, takes it directly to the patient, and administers it without a break in the process, the medication must be labeled.
                (2) Medication or solution labels should include name, strength, concentration, diluent and volume, and expiration date/time.
            (b) Reduce the likelihood of patient harm associated with the use of anticoagulant therapy

(c) Record and pass along correct information about a patient's medications at admission, transfer, and discharge. Find out what medications the patient is taking. Compare these medications with new medications given to the patient. Make sure patients know which medications to take when they are at home. Tell patients it is important to bring their up-to-date list of medications each time they visit a physician.

iv. Use alarms safely. Make improvements to ensure alarms on medical equipment are heard and responded to on time.

v. Prevent infection.
   (a) Use the hand cleaning guidelines from the Centers for Disease Control and Prevention or the World Health Organization. Set goals for improving hand cleaning. Use the goals to improve hand cleaning.

vi. Identify patient safety risks: Reduce the risk of suicide.

vii. Prevent mistakes in surgery using pre-procedural checklist:
   (a) Make sure that the correct surgery is done on the correct patient and at the correct place on the patient's body.
   (b) Mark the correct place on the patient's body where the surgery is to be done.
   (c) Have a time-out before the surgery to make sure that a mistake is not being made and document the time-out.

e. Chart-abstracted measures for hospitals. Note that beginning with 01/01/2020 discharges, the emergency department (ED), immunization (IMM), substance use (SUB), tobacco treatment (TOB), and venous thromboembolism (VTE) measures previously in the aligned CMS/TJC specifications manual have been moved to the *Specifications Manual for Joint Commission National Quality Measures* for hospital use.

i. Chart-abstracted measures from the hospital inpatient quality reporting (IQR) program; however, the hospital may still voluntarily report (critical care-focused measures)
   (a) Venous thromboembolism - VTE-6-Hospital-acquired, potentially preventable venous thromboembolism
   (b) Immunization - IMM-2 influenza immunization
   (c) Surgical Care Improvement Project - SCIP-Inf-4 cardiac surgery patients with controlled postoperative blood glucose

ii. Required chart-abstracted core measures from the *Specifications Manual for Joint Commission National Quality Measures*, version 2022B2, posted 9/6/2022 (critical care–focused measures)
   (a) Comprehensive stroke - CSTK-04 Procoagulant reversal agent initiation for intracerebral hemorrhage (ICH)
   (b) Comprehensive stroke - CSTK-05a Hemorrhage transformation for patients treated with intravenous alteplase therapy only
   (c) Comprehensive stroke - CSTK-05b Hemorrhage transformation for patients treated with intravenous alteplase therapy or mechanical endovascular reperfusion therapy
   (d) Comprehensive stroke - CSTK-06 Nimodipine treatment administered
   (e) Comprehensive stroke - CSTK-08 Thrombolysis in cerebral infarction (TICI) post-treatment reperfusion grade
   (f) Emergency department - ED-1a - Median time from ED arrival to ED departure for admitted ED patients
   (g) Emergency department - ED-2a - Admit decision time to ED departure time for admitted patients
   (h) Stroke - STK-01 Venous thromboembolism (VTE) prophylaxis
   (i) Stroke - STK-02 Discharged on antithrombotic therapy
   (j) Stroke - STK-03 Anticoagulation therapy for atrial fibrillation/flutter

(k) Stroke - STK-04 thrombolytic therapy
(l) Stroke - STK-05 antithrombotic therapy by end of hospital day 2
(m) Stroke - STK-06 discharged on statin medication
(n) Stroke - STK-08 stroke education

6. Det Norske Veritas (DNV) – Established in 2008:
   a. One of three CMS-approved hospital accreditation organizations.
   b. Over 500 hospitals have converted from TJC to DNV.
   c. DNV does not set any standards for quality, but allows the hospital leadership to develop quality initiatives (e.g., marking the surgical site is a TJC standard).
   d. If the institution is accredited by the DNV, the hospital leadership establishes these standards.
7. Healthcare Facilities Accreditation Program (HFAP)
   a. One of three CMS-approved hospital accreditation organizations, the HFAP was originally created in 1945 to conduct objective review of services provided by osteopathic hospitals.
   b. Provides accreditation to all hospitals, ambulatory care/surgical facilities, mental health facilities, physical rehabilitation facilities, clinical laboratories and critical access hospitals. HFAP also provides certification reviews for primary stroke centers.
   c. Standards are written and cross-walk written tied directly to the Medicare requirements.
8. AHRQ (Agency for Healthcare Research and Quality) – The health services research arm of the U.S. Department of Health and Human Services. Sponsors the National Quality Measures Clearinghouse – A "public repository for evidence-based quality measures and measure sets"
9. Hospital Quality Alliance – Created in 2002
   a. Hospital Compare was created through the efforts of Medicare and the Hospital Quality Alliance
   b. 2004 – Hospitals could voluntarily report data on 10 "starter-set" quality performance measures and receive incentive payment.
   c. 2005 – Quality-of-care data expanded to 21 measures
   d. 2008 – Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS) – First national, standardized, publicly reported survey of patients' perspectives of hospital care
   e. Reporting initiatives sharpens the focus on QI.
10. CMS
    a. Mission – To "ensure effective up-to-date health care coverage and to promote quality care for beneficiaries"
    b. Quality initiative (established in 2001) empowers consumers with quality-of-care information and encourages providers to improve the quality of care.
    c. Hospital quality initiative (established in 2003) – Hospitals must submit data measures or accept a reduction in payment.
       i. Pay-for-performance measures associated with quality measures
       ii. CMS has removed many previously reportable core measures; however, the institution can choose which previously required measures to report.
       iii. FY 2023 Payment Determination Hospital IQR Program Measures (selected measures)

Table 1. Selected CMS Hospital IQR Program Measures for FY 2024 Payment Update

Topic	Measure
Stroke Measure (STK) Set	
	• Discharged on antithrombotic therapy (STK-02) • Anticoagulation therapy for atrial fibrillation/flutter (STK-03) • Antithrombotic therapy by the end of hospital day 2 (STK-05) • Discharged on statin medication (STK-06)
Venous Thromboembolism (VTE) Measure Set	
	• Venous thromboembolism prophylaxis (VTE-1) • Intensive Care Unit Venous Thromboembolism Prophylaxis (VTE-2)
Sepsis Measure	
	• Severe sepsis and septic shock: Management bundle (Composite Measure)
Mortality Measures	
	• Hospital 30-Day, All-cause, Risk-Standardized Mortality Rate following Acute Ischemic Stroke (MORT-30-STK)
Readmission Measures	
	• Excess Days in Acute Care after Hospitalization for Acute Myocardial Infarction (AMI Excess Days)
	• Excess Days in Acute Care after Hospitalization for Heart Failure (HF Excess Days)
	• Excess Days in Acute Care after Hospitalization for Pneumonia (PN Excess Days)
Safe Use of Opioids	
	• Safe Use of Opioids – Concurrent Prescribing
Emergency Department (ED) Throughput Measures	
	• Admit Decision Time to ED Departure Time for Admitted Patients (ED-2)

CMS = Centers for Medicare & Medicaid Services; FY = fiscal year; IQR = inpatient quality reporting.

Information from: Centers for Medicare & Medicaid Services (CMS). CMS Measures Inventory Tool. Available at https://qualitynet.cms.gov/files/63568a21a90aef0017847af3?filename=IQR_FY2024_CMS_Measures_v2.pdf.

11. Institute for Safe Medication Practices (ISMP) – Established 1975
    a. Nonprofit organization responsible for targeting medication error prevention and safe medication use; a certified patient safety organization
    b. Based on a nonpunitive approach and system-based solutions
    c. Five key areas of focus: Knowledge, analysis, education, collaboration, and communication
    d. Medication Errors Reporting Program – Practitioner self-reporting program
    e. ISMP Quarterly Action Agenda and Bi-weekly Safety Alerts
        i. Can review bi-weekly and conduct gap analysis to proactively identify at-risk safety concerns.
        ii. All bi-weekly newsletters are combined into a quarterly action agenda to review and identify areas for opportunity to improve safety in medication use processes.
    f. ISMP Targeted Medication Safety Best Practices for Hospitals (2022–2023)
        i. Worksheet can be used to perform a gap analysis of an institution's performance with medication safety issues that continue to cause fatal and harmful errors to patients.
        ii. The interactive worksheet can be found at https://www.ismp.org/resources/worksheet-ismp-targeted-medication-safety-best-practices-hospitals.

12. The Choosing Wisely initiative of the American Board of Internal Medicine Foundation is a collaboration of critical care societies including the American Association of Critical Care Nurses, American College of Chest Physicians, American Thoracic Society, and Society of Critical Care Medicine.
    a. Do not order diagnostic tests at regular intervals (e.g., every day), but in response to specific clinical questions.
    b. Do not transfuse red blood cells in hemodynamically stable, non-bleeding ICU patients with a hemoglobin concentration greater than 7 g/dL.
    c. Do not use parenteral nutrition in adequately nourished critically ill patients within the first 7 days of an ICU stay.
    d. Do not deeply sedate mechanically ventilated patients without a specific indication and without daily attempts to lighten sedation.
    e. Do not continue life support for patients at high risk of death or severely impaired functional recovery without offering patients and their families the alternative of care focused entirely on comfort.
    f. Do not continue antibiotic therapy without evidence of need.
    g. Do not delay mobilizing ICU patients.
    h. Do not provide care that is discordant with the patient's goals and values.
    i. Do not administer as-needed sedative antipsychotic or hypnotic medications to prevent and/or treat delirium and treating the underlying causes of delirium without first assessing for removing and treating the underlying causes of delirium and using non-pharmacological delirium prevention and treatment approaches.

Patient Case

3. You are reviewing compliance with the national patient safety goals in the perioperative setting specific to appropriate medication labeling. Through observation, you notice that there are inconsistencies in how propofol is drawn up and labled in the syringe, if labeled at all. In the majority of cases, the name of the drug is placed on the label. What other information should be included on the label of the syringe?
   A. Patient name, drug name, dose, and expiration date/time.
   B. Drug name, concentration, diluent if applicable, and expiration date/time.
   C. Patient name, drug name, drug dose, and concentration.
   D. Drug name, drug dose, concentration, and diluent if applicable.

D. The goal of the ASHP Practice Advancement Initiative (PAI), formerly known as the Pharmacy Practice Model Initiative, is to significantly advance the health of patients by supporting futuristic practice models that support the most effective use of pharmacists as direct patient care providers. The initiative aims to assist leaders and practitioners in creating a framework, determining services, identifying emerging technology, developing templates, and implementing change. The ASHP Pharmacy Practice Model Initiative started in 2010. See Figure 1 for the Practice Advancement Initiative 2030 recommendations.

The goal of PAI 2030 is to significantly advance the health and well-being of people by supporting patient-centered care delivery models that optimize the most effective use of pharmacists as direct patient care providers.

PRACTICE-FOCUSED RECOMMENDATIONS

Patient-Centered Care

A1. Pharmacists should collaborate with patients, families, and caregivers to ensure that treatment plans respect patients' beliefs, values, autonomy, and agency.

A2. The pharmacy workforce should lead medication reconciliation processes during care transitions (e.g., emergency department, upon admission and discharge, ambulatory care setting, community pharmacy, long term care).

A3. The pharmacy workforce should collaborate with patients, caregivers, payers, and healthcare professionals to establish consistent and sustainable models for seamless transitions of care.

A4. Pharmacist documentation related to patient care must be available to all members of the healthcare team, including patients, in all care settings.

A5. The pharmacy workforce should partner with patients and the interprofessional care team to identify, assess, and resolve barriers to medication access, adherence, and health literacy.

A6. Patients must have access to a pharmacist in all settings of care.

Pharmacist Role, Education, & Training

B1. All pharmacists should have an individualized continuing professional development plan.

B2. Pharmacists should leverage and expand their scope of practice, including prescribing, to optimize patient care.

B3. Pharmacists should participate in and assume key roles on emergency response teams.

Technology & Data Science

C1. Pharmacists should use health information technologies to advance their role in patient care and population health.

C2. Pharmacy practice leaders should foster the development and application of advanced analytics (e.g., machine learning and artificial intelligence) in activities such as risk assessment, monitoring performance metrics, identifying patients in need of pharmacist care, optimizing medication use, and business management.

C3. Pharmacy practice leaders should be engaged in assessing emerging patient care technologies (e.g., mobile applications, monitoring devices, digital wearables or ingestables, blockchain technology) to support optimal medication-use outcomes.

C4. The pharmacy workforce should be competent in health information technology (including but not limited to analytics, automation, and clinical applications of technology) with ongoing education and training embedded at all stages of career development.

Pharmacy Technician Role, Education, & Training

D1. Pharmacy technicians should participate in advanced roles in all practice settings to promote efficiency and improve access to patient care.

D2. Pharmacy technicians should have complete responsibility for advanced technical and supporting activities (e.g., order fulfillment, tech-check-tech, regulatory compliance, supply chain management, diversion prevention, revenue cycle management, patient assistance programs).

Leadership in Medication Use & Safety

E1. Pharmacists should advance the use of pharmacogenomic information for personalized medication treatment.

E2. Pharmacists should assume leadership roles in medication stewardship activities at the local, state, and national levels.

Figure 1. ASHP Practice Advancement Initiative 2030.

Used with permission from: American Society of Health-System Pharmacists (ASHP). PAI 2030 Recommendations. Available at https://www.ashp.org/Pharmacy-Practice/PAI/PAI-Recommendations.

ORGANIZATION-FOCUSED RECOMMENDATIONS

Patient-Centered Care

A7. The pharmacy workforce, in all care settings, must have access to complete patient medical records and related health information.

A8. The pharmacy enterprise should be integrated and modeled to provide patient-centered care across the continuum (e.g., home and outpatient infusion, specialty pharmacy, community pharmacy, acute care).

A9. The pharmacy workforce should lead medication education for patients and caregivers that optimize outcomes, including in care transitions.

A10. Pharmacists should play an active role in ensuring that ethical principles drive clinical and business decisions related to medication use.

A11. Health systems must provide 24/7 pharmacy services with advanced clinical capability.

A12. Health systems should support innovative models for providing a safe and appropriate level of pharmacy services for small and rural hospitals or other alternative practice settings.

A13. Pharmacy departments should take responsibility for appropriate medication use in the structuring of external partnerships.

Pharmacist Role, Education, & Training

B4. Health systems should require completion of ASHP-accredited residency training as a minimum credential for new pharmacist practitioners.

B5. Pharmacists should participate in organization-based credentialing and privileging processes to ensure competency within their scope of practice.

B6. Pharmacy practice leaders should ensure that their workforce has the necessary knowledge and competency to adapt to emerging healthcare needs.

B7. Pharmacists practicing in specialty areas should be board-certified through the Board of Pharmacy Specialties or other appropriate body.

Technology & Data Science

C5. Virtual pharmacy services (e.g., telepharmacy) should be deployed to optimize operational and clinical services that extend patient care services and enhance continuity of care.

C6. The pharmacy enterprise must have sufficient resources to develop, implement, and maintain technology-related medication-use safety standards.

C7. Pharmacy departments should have access to an analytics resource, such as a data scientist, to collect, aggregate, measure, visualize, and disseminate data related to the financial and clinical performance of pharmacists.

C8. Pharmacy departments should use technology to ensure the safe compounding of sterile products.

Pharmacy Technician Role, Education, & Training

D3. All newly hired technicians should have completed an ASHP/ACPE-accredited technician education and training program.

D4. Health systems should require technicians to be certified by the Pharmacy Technician Certification Board.

D5. Pharmacy departments should foster the development of professional career paths for pharmacy technicians.

Leadership in Medication Use & Safety

E3. Pharmacy must be an active and accountable partner in the financial stewardship (e.g., minimizing waste, using cost-effective therapies, managing the supply chain) of care delivered in all settings.

E4. Pharmacy practice leaders should ensure evidence-based medication use by continually analyzing and reporting use patterns and outcomes.

E5. Health systems should support interprofessional innovation centers designed to pursue breakthroughs in areas such as patient experience, medication use, clinical outcomes, operational efficiency, technology, and revenue generation.

E6. Health systems should support the well-being and resiliency of their staffs.

E7. Pharmacy departments should strive to achieve equity, diversity, and inclusion in all technical, clinical, and leadership roles.

E8. The pharmacy enterprise should engage, employ, or develop expertise in areas such as finance, analytics, business management, quality assurance, informatics, human resources, payer relations, and supply chain management.

E9. Health systems should have a pharmacist executive leader, with a reporting structure consistent with other executive leaders, to oversee and influence enterprise-wide decision making related to medication use and technology.

E10. The pharmacy workforce should assess and mitigate risk in medication-use systems across all settings.

Figure 1. ASHP Practice Advancement Initiative 2030. (*continued*)

Used with permission from: American Society of Health-System Pharmacists (ASHP). PAI 2030 Recommendations. Available at https://www.ashp.org/Pharmacy-Practice/PAI/PAI-Recommendations.

PROFESSION-FOCUSED RECOMMENDATIONS

Patient-Centered Care

A14. Pharmacists should lead and advocate for comprehensive medication management in all healthcare settings.

Pharmacist Role, Education, & Training

B8. Pharmacy education, residency training, and continuing education should cover healthcare reimbursement, payment, and business management in all areas of practice.

B9. Pharmacists in all care settings should be included as integral members of the healthcare team and share accountability for patient outcomes and population health.

B10. The pharmacy workforce should be knowledgeable and have the resources to care for patients with behavioral and mental health disorders.

B11. Training and credentialing should exist to develop and recognize pharmacists who specialize in health information technology.

B12. Credentialed ambulatory-care pharmacists should be considered primary care providers.

B13. The profession should champion multi-state or national licensure for pharmacists.

B14. Pharmacists, in collaboration with other key stakeholders, must work to increase public, regulatory, and health professional understanding of pharmacists' roles and value.

Technology & Data Science

C9. Pharmacy should employ high-reliability principles when designing and selecting health information technology.

C10. Pharmacy should advocate for information technology that is interoperable and transparent with respect to usability, security, and functionality across the continuum of care.

C11. Pharmacy should establish standards for the application of artificial intelligence (AI) in the various steps of the medication-use process, including prescribing, reviewing medication orders, and assessing medication-use patterns in populations.

Pharmacy Technician Role, Education, & Training

D6. A scope of practice including core competencies should be developed and defined for pharmacy technicians in acute-care and ambulatory-care settings.

Leadership in Medication Use & Safety

E11. Pharmacists should lead the development, implementation, and evaluation of medication-related national quality indicators and accountability measures.

E12. Pharmacists should be leaders in federal and state legislative and regulatory policy development related to improving individual and population health outcomes.

E13. Pharmacy should partner with interprofessional organizations to define and delineate practice advances into state and federal laws and regulations to optimize patient care.

E14. Pharmacy should leverage healthcare models that acknowledge pharmacist value and align payment with quality of outcomes.

Figure 1. ASHP Practice Advancement Initiative 2030. (*continued*)

Used with permission from: American Society of Health-System Pharmacists (ASHP). PAI 2030 Recommendations. Available at https://www.ashp.org/Pharmacy-Practice/PAI/PAI-Recommendations.

E. Local Quality Initiatives (from ASHP: The ASHP Discussion Guide on the Pharmacist's Role in Quality Improvement)
   1. Every accredited hospital must participate in QI initiatives.
   2. The overall goals of the PI programs are to maintain and support the delivery of safe, quality care.
   3. Each program should include the following:
      a. Adherence to standards of care
      b. Opportunities for improvement, with action plans to implement change strategies
      c. Strategies for the effectiveness of change strategies
      d. Involvement of multidisciplinary teams in process improvement
   4. PI plans should:
      a. Articulate the commitment to PI
      b. Delineate the goals of the PI process
      c. Specify the authority and responsibilities for PI
      d. Describe the organizational structure and process related to the PI program
      e. Describe the method for improving organizational performance
      f. Describe the communication and recognition of PI activities

F. QI Tools
   1. Six Sigma – DMAIC process: Goal to reduce variation around process defects and test with statistical methods
      a. DEFINE the problem
      b. MEASURE current state through process mapping, or Failure Modes and Effects Analysis (FMEA)
      c. ANALYZE relationship between input/output
      d. IMPROVE future state processes
      e. CONTROL to sustain the gain
   2. Failure Modes and Effects Analysis (FMEA): Prospective evaluation to determine possible failures in processes in order to prevent them from occurring on initiation
      a. Evaluate the steps in the process
      b. What could go wrong? (Failure modes)
      c. Why would the failure happen? (Failure causes)
      d. What would be the consequences of each failure? (Failure effects)
   3. LEAN principles: How do we do more with less? Ideally, create a high-quality service/product with less resources or a shorter lead time
      a. Value - Activity that enhances service from the customer's viewpoint
         i. Waste – opposite of value with goal to eliminate waste
         ii. Types of waste: Defects, Overproduction, Waiting, Not using employees' creativity, Transport, Inventory, Motion, Extra Processing ("DOWNTIME")
      b. Value stream – all actions required to bring service to the customer (both value and non-value added)
         i. Value stream mapping is a LEAN tool that can be used to "walk the line" by experiencing the process and defining the current state.
      c. Flow – flow of processes without interruption
         i. Process mapping to outline the workflow
      d. Pull – trigger flow from demand
      e. Perfection – continuous improvement

4. Plan-do-study-act (PDSA): Model for promoting quality improvement through real-time evaluation of standard work
    a. What are we trying to accomplish? – An aim or project goal must be developed (example of specific aim: time to appropriate antibiotic therapy in patients seen in the ED with presumed sepsis).
        i. Describe the process to be improved.
        ii. Set a target for improvement that extends beyond current performance.
        iii. Secure necessary resources in the process.
    b. How will we know that a change is an improvement?
        i. Establish baseline measurement – Select and gather data.
        ii. Compare baseline data with the target and identify key causes and sources of variation.
    c. What changes can we make that will result in improvement?
        i. Broad, general ideas and thoughts about change ("change concepts")
        ii. Select the change.
    d. Testing change
        i. Scientific method
    e. Plan-do-study-act cycle: Use an A3, which is a visual management system to identify opportunities for improvement to ensure they align with system goals. A3 was developed by Toyota to communicate with as few words as possible.
        i. Plan – Analyze the problem or opportunity and search for the root cause. Design the process improvement and implementation plan.
            (a) Questions are asked and predictions are made.
            (b) Details are described.
            (c) Who will make the test?
            (d) What exactly will they do?
            (e) When will they do it?
            (f) Where will they do it?
            (g) How long will they do it?
            (h) Predict what will happen.
            (i) Determine which data to collect.
        ii. Do – try it out
            (a) Change is tested following the "plan."
            (b) Collect data on the change.
            (c) Unexpected problems and observations are documented.
        iii. Study – analyze data, study results
            (a) Study the effect of the test change on the single measure.
            (b) Compare data with predictions.
            (c) Summarize what was learned.
        iv. Act – refine the change on the basis of what was learned
            (a) Adapt, adopt, or abandon change idea for next cycle.
            (b) Develop an implementation plan.
            (c) Share final reflections.

G. Data Generation
    1. Benchmarking – Compares your performance with that of another health care system
    2. Provider concerns
    3. Nursing reports/concerns
    4. Patient concerns or comments: Reports from third-party payers and regulatory agencies
    5. Incident reporting
    6. Root-cause analysis reports

7. Medication error reports
8. Patient care evaluation studies
9. Patient satisfaction survey results
10. CPOE use reports
11. Bar code medication administration data reports
12. Continuous infusion pump integration data reports

H. Analysis Tool Kit
1. Affinity diagrams
2. Cause-and-effect diagrams
3. Decision matrixes
4. Root-cause analysis
5. Apparent cause analysis
6. Process maps
7. Run chart, control chart, or flowcharts
8. Force-field analysis
9. Histograms, scatterplots
10. Relationship diagrams
11. Supplier, input, process, output, customer
12. Voice of the customer
13. Five Whys
14. Fishbone diagram
15. FMEA

I. Publishing QI Results
1. Important to publish QI data to promote improvement efforts in health care
2. Reduces the chances of the same mistakes being repeated
3. Shares your work by spreading important and useful information
4. Standards for Quality Improvement Reporting Excellence (SQUIRE) – Provides a framework for reporting new knowledge about how to improve health care. Available at www.squire-statement.org/

Patient Cases

4. As a critical care pharmacist, you are involved in a QI program. Which TJC measure is best known to be affected by the critical care pharmacist?
   A. ED median time from ED arrival to ED departure for admitted ED patients.
   B. Procoagulant reversal agent initiation for intracerebral hemorrhage (ICH).
   C. Hemorrhagic transformation for patients receiving intravenous alteplase therapy or mechanical endovascular reperfusion therapy.
   D. Severe sepsis and septic shock management bundle.

5. As part of a QI initiative, you identify an increase in the monthly rate of catheter-related bloodstream infections in your ICU. You decide to evaluate the current practice using LEAN methodology. Which is the best example of a component of LEAN?
   A. Define the problem.
   B. Ask your nurses why they think a spike in catheter-related bloodstream infections has occurred.
   C. Identify transportation barriers.
   D. Describe the past process.

IV. MEDICATION USE EVALUATION

A. TJC requires medication use evaluations (MUEs) to be completed to monitor the safety of medications. ASHP published guidelines on performing a MUE in 2021.

B. MUEs and DUEs are often used synonymously. Both DUEs and MUEs are PI and QA methods that ensure optimal medication therapy management and improve patient safety and outcomes.

C. A DUE is drug- or disease-specific, focusing specifically on the criteria-based assessment of the medication use process and prescribing, whereas an MUE provides a broader scope that focuses on a drug or class of drugs, the process(es), and the outcome, with a specific emphasis on improving patient outcomes. MUEs focus on several elements of the medication process/use such as prescribing, pharmacist medication order validating or verifying, dispensing, preparing, administering, monitoring, patient education, and outcomes.

D. ASHP Guidelines on MUEs: Include a performance improvement framework to improve safety, efficacy, quality, and efficiency in patient care.
   1. An example of a performance improvement framework is FOCUS-PDSA:
      a. Find the process to be targeted for improvement.
      b. Organize the team that knows the process.
      c. Clarify current knowledge of the process.
      d. Understand causes of process variation.
         i. Five Whys
      e. Select process improvement.
         i. FACES (feasibility, acceptability, cost/benefit, effectiveness, sustainability) tool
      f. Plan: Develop a solution.
      g. Do: Implement improvements.
      h. Study: Evaluate the results.
      i. Act: Determine the changes needed moving forward and implement them.

E. The sample size of the MUE depends on the type of medication data being analyzed—usually, a sample of 20–30 patients is sufficient; however, a sample of 100 or more patients may be required to analyze patient outcomes.

F. Data collection for specific criteria can use either a yes/no format (with a section for comments) or open-ended questions. Using CPOE and electronic medical records, the processes of data retrieval, monitoring, and generating specialized reports have become easier.

G. The type and number of MUEs should be determined by the risk mitigated when using a medication. Broad categories for MUEs include:
   1. Promote optimal medication therapy
   2. Improve patient safety
   3. Standardize to reduce unnecessary variation
   4. Optimize drug therapy
   5. Assess value of innovative practices
   6. Meet quality or regulatory standards
   7. Minimize costs
      a. These should be considered together with the impact of the medication or medication use process on patients and the medication use system as a whole to prioritize the highest-impact MUEs.

H. Medications selected for an MUE may be based on the following:
   1. High risk
   2. High volume
   3. ADEs
   4. Preventable ADEs
   5. Near-miss and harmful medication errors
   6. Formulary review, including addition (non-formulary use), retention, or deletion
   7. Pharmacy intervention data
   8. Treatment failures
   9. Physician or nurse identification or request
   10. Patient concerns
   11. Off-label use
   12. High cost
   13. Beneficial when used a specific way
   14. Drug shortages, recalls, or safety alerts

I. Examples of medication use processes in critically ill patients that may be selected for an MUE can be found in Box 1.

Box 1. Examples of High-Risk Medication-Related Processes Suited for an MUE in Critically Ill Patients

PADIS (ABCDEF bundle) protocols
Use of neuromuscular blocking agents
Alcohol withdrawal management
Management of supratherapeutic anticoagulation (e.g., warfarin, direct oral anticoagulants, heparin, argatroban)
Use of β-blockers in myocardial infarction
Monitoring for dysrhythmias with QTc-prolonging drugs
Antihypertensive IV-to-PO switch therapy
Antihypertensive use in acute stroke
Fluid resuscitation
Management of gastrointestinal bleeding
Use of total parenteral nutrition
Use of albumin
Hyperkalemia management guidelines
Hypomagnesemia management guidelines
Management of status epilepticus
Management of hyponatremia
Use of IV sodium bicarbonate
Vancomycin dosing and ordering serum concentrations
Aminoglycoside dosing and ordering serum concentrations
Intermittent infusions of antimicrobials (e.g., carbapenems, piperacillin/tazobactam)
Evaluation of antimicrobial dosing in CRRT or hemodialysis
Surgical prophylaxis guidelines
Antimicrobial IV-to-PO switch therapy
Management of *Clostridioides difficile* diarrhea
Vaccine administration
IV push medication guidelines (rate of administration and medication preparation - diluted or undiluted)
Stress ulcer prophylaxis
Insulin infusions
Hypoglycemic protocols

CRRT = continuous renal replacement therapy; PO = oral(ly); QTc = corrected QT (interval).

J. Ideally, MUEs should be performed proactively and used to identify any gaps in practice or patient safety.
   1. An *interventional MUE* is completed concurrently or prospectively, and if the criteria are not met, an intervention by the pharmacist should be made to improve the use of the medication and patient outcomes.
   2. A *noninterventional MUE* is completed concurrently or retrospectively by a medical record review, and data are collected but criteria are not met. The review is retrospective, and there is no pharmacist-to-prescriber interaction during the review process.

K. MUEs should have specific criteria set – These criteria are best determined by a multidisciplinary team of medication or disease experts.

L. MUE criteria may be approved by the P&T committee. The P&T committee may create an MUE subcommittee. Other hospital committees such as the QI committee may also request that an MUE be performed. Other names for the MUE committees may include formulary, drug safety, therapeutic assessment, medication safety, and drug use review committees.

M. The MUE subcommittee should be multidisciplinary and may be composed of the following:
   1. Clinical pharmacists
   2. Physicians, nurse practitioners, and physician assistants
   3. Nurses
   4. Administrators
   5. PI/QA representatives
   6. Risk management representatives
   7. Respiratory therapists, nutritionists, and other ancillary support staff

N. Given the pharmacist's expertise in medication management, pharmacists often chair or co-chair the MUE subcommittee and perform the MUE.

O. The MUE subcommittee can recommend drugs and drug processes that require an MUE to the P&T committee; alternatively, the P&T committee can request MUEs from the MUE subcommittee.

P. The MUE process includes reviewing the findings and developing plans of improvement. The results and conclusions of the MUE should be reported to the P&T committee and department chairs.

Q. After plans of for improvement are identified and actions taken, a follow-up MUE should be completed to document that improvement has occurred.

R. Periodic MUEs in the same area should be performed every 3–6 months for 1 year to ensure that the improvements made remain effective and are sustained. If any new changes have occurred in the medication use process, the MUE criteria should be reassessed and the new criteria incorporated.

S. Policies and educational initiatives should be developed to improve opportunities identified from the MUE, such as:
   1. In-service lectures
   2. Newsletter publications
   3. Medication alerts
   4. Guideline development
   5. Protocols
   6. Policy and procedures
   7. CPOE pathways, prescribing guides, or information or pop-up warnings

Patient Case

6. Which medication use process is best suited for an MUE?
   A. Review of pharmacist notes in the electronic health record (EHR).
   B. Review of accuracy of expiration dates placed on intravenous admixture products.
   C. Review of vancomycin dosing and ordering of vancomycin blood concentrations.
   D. Review of frequency of drug interaction warnings on the CPOE system.

V. EDUCATION

A. Educate health care professionals and other stakeholders concerning issues related to the care of critically ill patients.
   1. Provides formal and informal instruction to pharmacists and other ICU health care professionals
   2. Participates in the training of pharmacy and medical students, residents, and fellows through experiential critical care rotations
   3. Provides critical care pharmacology and therapeutic didactic lectures to health care professionals, including students, residents, and fellows
   4. Implements pharmacist and pharmacy technician training programs for ICU personnel
   5. Provides accredited continuing education sessions
   6. Develops educational materials such as pocket guides or tip sheets related to the care of critically ill patients
   7. Educates lay group community medical personal about the role of the ICU pharmacist
   8. Coordinates or directs internships; experiential training; traineeships, residency, or fellowship programs
   9. Teaches advanced cardiac life support

B. Communication Strategies
   1. The method of delivery may influence the degree of learning and retention among the target audience.
      a. "In person" education is generally more effective than email/written communication/education.
   2. Evaluate the audience.
      a. Patient/general public
      b. Health care providers
      c. Learners (e.g., students, residents)
      d. Legislators
      e. The media

3. Communicate (see Table 2).

Table 2. Communication Strategies

Strategy	Questions to ask regarding the communication
Credibility	Is your messenger credible? Is the messenger a trusted and respected source of information with your audience?
Context	Is your message in context with reality and the environment in which your audience is located?
Content	Is your message relevant to your audience? Is the audience interested in the information?
Clarity	Is your message straightforward? How far will it travel and how long will it last? Do not use abbreviations
Continuity and consistency	Repeat your message for audience penetration
Channels	What channels/tools of communication are you using? What value are they bringing to your audience?
Customer benefits	What is in it for me?
Caring, compassion, and concern	Does your audience know that you care?
Capability of audience	Is your audience capable of understanding the message? Will the audience take the time to read, watch, or listen to it?
Call to action	What is your audience supposed to do now?

4. Strategies and tools to communicate the message or information:
   a. Pharmacy departmental newsletter
   b. Hospital newsletter
   c. Electronic screensavers providing information
   d. Best practice advisory methods
   e. Local/regional communication newsletter/e-mail list
   f. Electronic message such as integration into the EHR if possible
   g. Education must be an ongoing process and often must be repeated over time to capture personnel changes and/or drift in knowledge/behaviors.
   h. "Grand Rounds" or other live or virtual, formal presentations
   i. Informal presentations at daily "huddles"
   j. Presentations or approval at committee level meetings
   k. Presentations at local and national professional society meetings (live or virtual)

VI. DOCUMENTATION PROCESSES USED FOR CRITICAL CARE PHARMACY SERVICES

A. Importance of Documentation
   1. Evidence of economic benefit and improvement in patient safety for clinical pharmacy services (CPS) and critical care pharmacy services is well established. The Society of Critical Care Medicine supports an ICU pharmacist as an essential component of the ICU team.
   2. Clinical pharmacy interventions that affect patient care should be documented in the electronic medical record. Pharmacist involvement directly optimizes the use of medications, decreases drug-related costs, prevents adverse effects, improves quality and efficacy of care, reduces mortality, shortens length of stay (LOS), and lowers overall patient care costs.

3. Clinical interventions that can be performed by a critical care pharmacist may include but are not limited to:
    a. Optimization of the correct drug for the disease process (inappropriate therapy, duplication of therapy, therapy no longer needed, etc.)
    b. Optimization of the correct dose based on the patient's end organ function (renal dose adjustment, hepatic dose adjustment)
    c. Anticoagulation adjustment
    d. Antibiotic optimization (deescalation, bug-drug mismatch, duration of therapy, etc.)
    e. Intravenous to oral conversion
    f. Transitions of care and/or medication reconciliation
    g. Pharmacokinetic dosing
    h. Adverse event management
    i. Drug-drug or drug-disease management
    j. Ensuring adherence to ICU medication therapy protocols (i.e., alcohol withdrawal, PADIS management, neuromuscular blockade, etc.)
    k. Drug and laboratory monitoring
4. Additionally, critical care pharmacist interventions may identify opportunities for protocol or guideline development or may identify opportunities for an MUE to be completed.
5. Documenting CPS is important not only to quantify CPS and workload, but also to provide justification for maintaining and expanding services.
6. CPS should be evaluated for cost impact and cost outcomes savings and should include the following:
    a. Weighted metric for each variable to quantify measured activities
        i. One way to add impact to each intervention is by using the Overhage and Lukes Scale to capture severity of interventions.
    b. Pharmacotherapy improvement
    c. Cost savings or cost avoidance
    d. Antibiotic stewardship
    e. Provider education
    f. Quality/safety improvement (value-based purchasing, HCAHPS)
    g. Participation in emergency response (cardiac arrest, stroke, sepsis)
    h. Prospective chart review
    i. Rounding with health care providers
    j. Formal pharmacy consults
7. Collaborative drug therapy management (CDTM) or collaborative practice agreements (CPAs) – When a prescriber and a pharmacist establish written guidelines or protocols authorizing the pharmacist to initiate, modify, or continue drug therapy for a specific patient. CDTM is a type of pharmacotherapeutic intervention that should be documented and reported. Examples relevant to the practice of critical care pharmacy include parenteral nutrition prescribing, antimicrobial stewardship programs, and analgesia/sedation/delirium screening and management.

Patient Case

7. You round daily in your ICU as part of the multidisciplinary team. Your manager asks you to help them justify clinical pharmacy services to obtain additional FTEs for the other ICUs at your institution. Which of the following would be the most effective metrics to include in the justification to hospital leadership?
    A. Antibiotic stewardship and cost savings.
    B. Provider education and in-services provided.
    C. Time spent rounding and on other activities.
    D. Renal dose adjustments and intravenous to oral conversion.

B. Documentation Process
   1. The pharmacy department should have a policy and procedure for documenting clinical pharmacy interventions. In addition, a dashboard should be developed that incorporates these clinical interventions (Appendix 2).
   2. In general, pharmacist interventions are the result of either formal or informal consultations or unsolicited interventions.
   3. Documenting the pharmacotherapeutic intervention and plan in the EHR provides transparency between all health care professionals – physicians, nurses, pharmacists, dietitians, respiratory therapists, and social workers and provides written documentation of recommended changes.
   4. Pharmacists can also document interventions in the pharmacy profile; however, this method generally allows for review only by pharmacy personnel who have access to the pharmacy computer system such as other pharmacists and pharmacy interns, students, and technicians.
   5. Both methods should be available for documenting pharmacotherapeutic interventions, and criteria should be established to determine which method is most appropriate.
   6. Pharmacists should have the authority to document pharmacotherapeutic interventions in the EHR.
   7. Pharmacists should be trained and educated to document in the EHR. The ASHP Clinical Skills Certificate program is a tool that can be used to train pharmacists to document in the EHR.
   8. Documentation methods may include standard format documentation methods such as:
      a. SOAP (subjective, objective, assessment, plan)
      b. TITRS (title, introduction, text, recommendation, signature)
      c. FARM (findings, assessment, resolution, monitoring)
   9. Wording for solicited consultations should be direct. Unsolicited pharmacist interventions should be documented subtly, allowing the primary provider to decline the recommendation without incurring liability. Phrases that can be used include:
      a. "May consider"
      b. "Suggest"
      c. "May recommend"
   10. When feasible, written notes by pharmacists should be documented in the EHR after an oral communication with the clinician; this allows for any patient data discrepancies to be corrected and for agreement and confirmation between the prescriber and the pharmacist to execute the intervention.
   11. Pharmacists should follow up on their patient interventions daily and provide follow-up notes that include patient progress or new interventions, when needed.
      a. Pharmacists should provide their contact information.
      b. Co-signatures should be required for pharmacy residents (until deemed competent according to pharmacy department standards), interns, and students.
      c. Continual physical presence during and after direct patient care rounds should be provided to support the physicians, advanced practice providers, and nursing team with triaging medication-related questions.
   12. Many web-based or handheld electronic systems are available that can be used to document and report pharmacotherapeutic interventions and cost savings. Ideally, these are documented in the patient's EHR for the reporting of weighted metrics.

Patient Case

8. During ICU rounds, the team is recommending a cost-prohibitive non-formulary drug with controversial outcomes as a last line vasopressor for a patient with septic shock. You review the chart and think there are safer formulary options to try before resorting to this new therapy. You are required to document your interventions in the patient chart for tracking purposes. Which wording is the most appropriate to use?
   A. Because of the significant cost of this drug, the team should resort to other formulary options.
   B. This drug was associated with significant adverse effects in clinical trials and will likely cause harm if used in this patient.
   C. The team should consider optimizing the patient's pharmacological therapy with agents that are on formulary and have a proven benefit in patients with septic shock.
   D. The use of this non-formulary drug would be a poor choice and likely lead to patient harm if used.

C. Evaluation of CPS
   1. Continuous QI should include quality indicators and periodic reviews of pharmacist-written documentation and consultations.
   2. These reporting systems allow for collecting, aggregating, and benchmarking data against data from other hospitals and bed size. They increase the credibility of data collection methods and results when evaluated by health care administrators (Appendix 2).
   3. To document raw drug cost savings from changing to less expensive medications, the following method may be applied: subtract the cost of the originally prescribed drug therapy (drug daily cost multiplied by the number of days prescribed) from the cost of the less expensive drug therapy (drug daily cost multiplied by the number of days prescribed). Using this method, the cost of intravenous diluents and admixture fluids and syringes used in the preparation process may be included. Medication costs can also be affected by factors such as shortages and FDA granting using a New Drug Application.
   4. Value of pharmacist interventions should be determined as well. Value may be assigned utilizing cost-avoidance associated with types of pharmacist interventions or with the Overhage and Lukes scale.
   5. Documentation of interventions for reporting to other hospital committees such as the P&T committee should include the following:
      a. Date, time
      b. Type of intervention
      c. Drug(s) involved
      d. Prescriber name, service, and type of health care provider
      e. Duration of time spent completing the intervention
      f. Whether the intervention was accepted or denied, or clarification was achieved
      g. Value of the intervention (e.g. cost avoidance, Overhage & Lukes scale)
   6. Documentation of services should show diversity, effectiveness, cost, and outcomes of activities.
      a. Policy development
      b. Research
      c. Resource use
      d. Management
      e. Leadership
      f. Education
      g. Health care professionals (e.g., physicians, nurses, respiratory therapists)
         i. Pharmacy students
         ii. Pharmacy residents (PGY1, PGY2) and pharmacy fellows
         iii. Pharmacy personnel

7. Outcomes of documentation
    a. Establish additional clinical services.
    b. Expand roles of existing services.
    c. Assess new processes or practices (prescriber privileges or provider reimbursement).
    d. Provide data for QA or research initiatives.
    e. Accreditation purposes
    f. Promotional reasons
    g. Financial impact can be analyzed and used to justify time spent in that area.
    h. Generate a business plan to expand clinical services:
        i. Background and description
        ii. Market and competitor analysis
        iii. Operational structure and processes
        iv. Financial projections
        v. Milestones, schedules, and action plan
        vi. References
        vii. Supportive documents
        viii. Financial pro forma statements
        ix. Letters of support
8. Reports
    a. Statistical interpretations of services
    b. Satisfaction surveys of their services
    c. Risk reduction
    d. Peer review
    e. Publication
    f. Return on investment data
    g. Future initiatives
9. Pharmacotherapeutic intervention data should be presented and emphasized at as many committee opportunities as possible – this increases the visibility and corroborates the importance of the pharmacy department and the critical care pharmacy services. Box 2 lists examples of pharmacotherapeutic interventions that can be reported. Pharmacotherapeutic intervention data should be presented to the following:
    a. ICU committee
    b. P&T committee
    c. QA committee
    d. Use review committee
    e. Medical executive committee – Engages with potential expansion of pharmacy services or cost avoidance
10. Pharmacotherapeutic intervention data may be presented quarterly (Figure 2). These data should include the type of intervention (Figure 3) and both the raw drug cost savings derived from changing to less expensive drugs and the outcome cost savings such as preventing an adverse event and decreasing the LOS (Figure 4).

Box 2. Examples of Pharmacotherapeutic Interventions

1. Discontinue drug 2. Switch drug 3. Switch drug for less expensive but equally safe and effective drug 4. Drug dosing 5. Drug dosage form 6. Drug route 7. Pharmacokinetic consultation 8. Pharmacotherapy consultation 9. IV-to-PO switch therapy 10. Contraindication 11. Duplicate therapy 12. Nonformulary switch to formulary 13. Nonformulary approval 14. Drug unavailable 15. IV drug incompatibility 16. Drug-drug interaction 17. Drug-food interaction 18. Drug-laboratory test interaction 19. Pharmacogenomics interaction 20. Therapeutic interchange 21. Allergy prevention	22. Medication reconciliation 23. Transfer medication reconciliation 24. Discharge medication reconciliation 25. Discharge counseling 26. Written patient education provided 27. Patient adherence to medications 28. Adverse drug reaction 29. Overdosage 30. Subtherapeutic dose 31. Medication error identification and avoidance 32. Monitoring for efficacy 33. Monitoring for toxicity 34. Ordering laboratory tests for monitoring drug efficacy and toxicity 35. Clarification of medication order 36. Untreated indication 37. Failure to receive medication 38. Immunization recommendation 39. Immunization administered 40. Health risk assessment 41. Rapid response interventions 42. Code interventions 43. CDTM interventions

CDTM = collaborative drug therapy management.

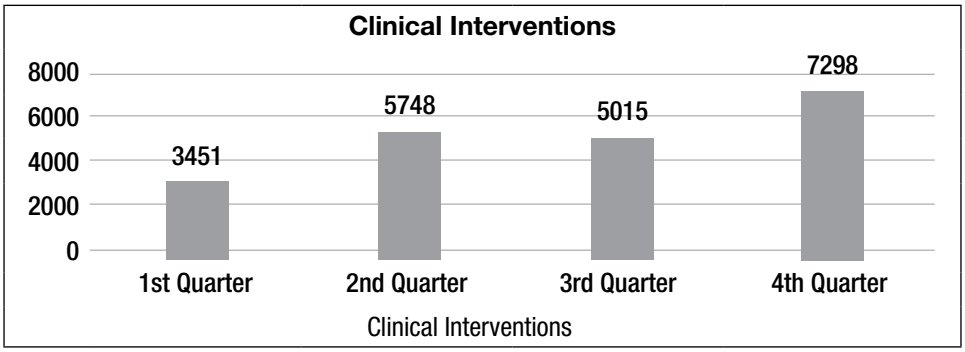


Figure 2. Clinical interventions.

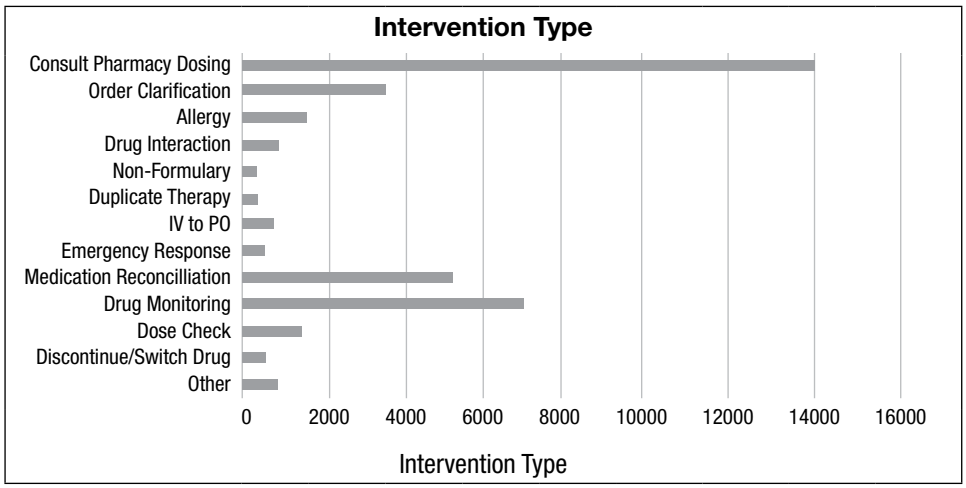


Figure 3. Pharmacotherapy intervention types.

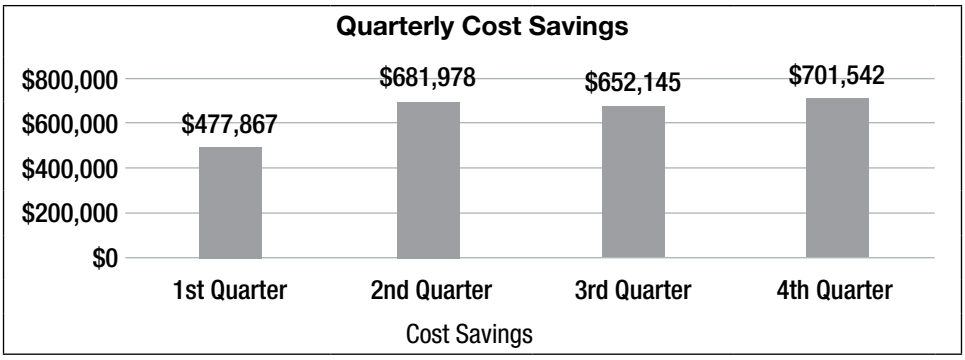


Figure 4. Raw and outcome drug cost savings

11. Benchmarking is important in annualizing the total number of pharmacotherapeutic interventions with previous years (Figure 5) and the cost savings with previous years (Figure 6) and in providing rational explanations for any discrepancies noted.

12. Change in the number of interventions and/or potential cost savings should be evaluated cautiously. Changes in number of pharmacists, implementation of guidelines/protocols to standardize care, etc., may impact these numbers and should be considered when assessing.

13. Pharmacotherapeutic interventions that are reported may be prioritized according to their clinical impact on patient safety and cost savings. Examples of high-priority pharmacotherapeutic interventions that should be reported include the following:
    1. Allergy prevented
    2. Contraindication prevented
    3. Drug dosing adjustments
    4. Duplicate therapy avoided
    5. Drug interactions avoided
    6. Medication reconciliation intervention
    7. Ordering laboratory tests for monitoring of drug safety and efficacy
    8. Switching drug for less expensive but equally safe and effective drug

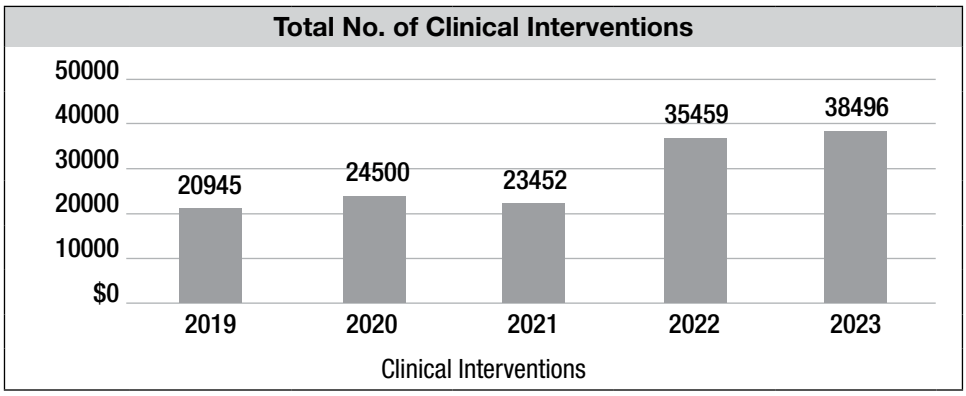


Figure 5. Annual pharmacotherapy interventions 2019–2023.

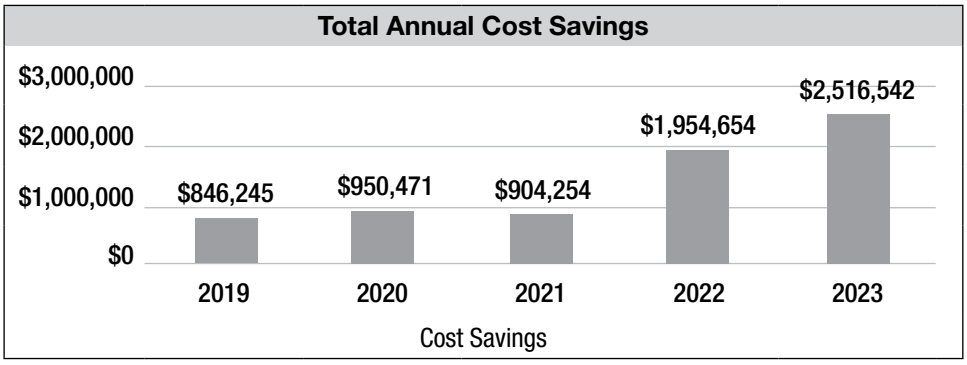


Figure 6. Annual raw drug and outcome cost savings 2019–2023.

REFERENCES

Policy, Procedure, and Guidelines

1. American Society of Health-System Pharmacists (ASHP). ASHP Guidelines on Medication Use Evaluation. Available at https://www.ashp.org/-/media/assets/pharmacy-informaticist/docs/sopit-formulary-guideline-medication-use-evalutation.ashx?la=en.
2. Ciccarello C, Leber MB, Leonard MC, et al. ASHP guidelines on the pharmacy and therapeutics committee and the formulary system. Am J Health Syst Pharm 2021;78:907-18. Available at www.ashp.org/-/media/assets/policy-guidelines/docs/guidelines/gdl-pharmacytherapeutics-committee-formulary-system.ashx.
3. Grol R, Grimshaw J. From best evidence to best practice: effective implementation of change in patients' care. Lancet 2003;362:1225-30.
4. Sackett DL, Rosenberg WMC, Gray JAM, et al. Evidence-based medicine: what it is and what it isn't. BMJ 1996;312:71.

Gap Analysis

1. ASHP Foundation. Pharmacy Forecast 2021: Strategic Planning Advice for Pharmacy Departments in Hospitals and Health Systems. Available at Am J Health Syst Pharm 2021;78:472-97.
2. Centers for Disease Control and Prevention (CDC). The Core Elements of Hospital Antibiotic Stewardship Programs 2019: Antibiotic Stewardship Program Assessment Tool. Available at https://www.cdc.gov/antibiotic-use/healthcare/pdfs/hospital-core-elements-H.pdf.
3. Devlin JW, Skrobik Y, Gelinas C, et al. Clinical practice guidelines for the prevention and management of pain, agitation/sedation, delirium, immobility, and sleep disruption in adult patients in the ICU. Crit Care Med 2018;46:e825-73.
4. Golden SH, Hager D, Gould LJ, et al. A gap analysis needs assessment tool to drive a care delivery and research agenda for integration of care and sharing of best practices across a health system. Jt Comm J Qual Patient Saf 2017;43:18-28.
5. Institute for Safe Medication Practices (ISMP). Targeted Medication Safety Best Practices for Hospitals (2022-2023).
6. Kane-Gill SL, Dasta JF, Buckley MS, et al. Clinical practice guideline: safe medication use in the ICU. Crit Care Med 2017;45:e877-e915.

Quality Improvement

1. American Society of Health-System Pharmacists (ASHP). Applying LEAN to the Medication Use Process. Available at https://www.ashp.org/-/media/assets/pharmacy-practice/resource-centers/quality-improvement/learn-about-quality-improvement-applying-lean-medication-use.ashx.
2. American Society of Health-System Pharmacists (ASHP). The ASHP Discussion Guide on the Pharmacist's Role in Quality Improvement. Available at ashp.org/-/media/assets/pharmacy-practice/resource-centers/leadership/leadership-of-profession-pharmacists-role-quality-improvement-guide
3. American Society of Health-System Pharmacists (ASHP). The consensus of the Pharmacy Practice Model Summit. Am J Health Syst Pharm 2011;68:1148-52.
4. American Society of Health-System Pharmacists (ASHP) Practice Advancement Initiative (PAI). 2030 Available at https://www.ashp.org/Pharmacy-Practice/PAI/About-PAI-2030.
5. Choosing Wisely. Five Things Physicians and Patients Should Question. Available at www.choosingwisely.org/clinician-lists/#keyword=critical_care.
6. Herring SR, Jones BR, Bailey BP. Idea generation techniques among creative professionals. In: Proceedings of the 42nd Hawaii International Conference on System Sciences – 2009. Available at http://orchid.cs.illinois.edu/publications/HICSS-idea-generation-2009.pdf.
7. Institute for Healthcare Improvement (IHI). Failure Modes and Effects Analysis (FMEA) Tool. Available at www.ihi.org/resources/Pages/Tools/FailureModesandEffectsAnalysisTool.aspx.
8. ISMP Targeted Medication Safety Best Practices for Hospitals (2020-2021). Available at https://www.ismp.org/resources/worksheet-ismp-targeted-medication-safety-best-practices-hospitals.

9. Joint Commission: Performance Measurement. Available at https://manual.jointcommission.org/pub/Manual/ReleaseNotesArchive/TJC_v2021B.pdf.
10. U.S. Department of Health and Human Resources. Quality Improvement. Available at https://www.hrsa.gov/sites/default/files/quality/toolbox/508pdfs/qualityimprovement.pdf.
11. Varkey P, Reller MK, Resar RK. Basics of quality improvement in health care. Mayo Clin Proc 2007;82:735-9.

Medication Use Evaluation

1. American Society of Health-System Pharmacists (ASHP). ASHP Guidelines on Medication-Use Evaluation. Available at https://www.ashp.org/-/media/assets/policy-guidelines/docs/guidelines/medication-use-evaluation-current.ashx.

Documentation of Pharmacy Services

1. Andrawis M, Ellison C, Riddle S, et al. Recommended quality measures for health-system pharmacy: 2019 update from the Pharmacy Accountability Measures Work Group. Am J Health Syst Pharm 2019;76:874-87.
2. Brilli RJ, Spevetz A, Branson RD, et al. Critical care delivery in the intensive care unit: defining clinical roles and the best practice model. Crit Care Med 2001;29:2007-19.
3. Cassat S, Massey L, Buckingham S, et al. Development of health-system inpatient pharmacy clinical metrics. Am J Health Syst Pharm 2019;76:1958-64.
4. Dager W, Bolesta S, Brophy G, et al. An opinion paper outlining recommendations for training, credentialing, and documenting and justifying critical care pharmacy services. Pharmacotherapy 2011;31:135e-75e.
5. Horn E, Jacobi J. The critical care clinical pharmacist: evolution of an essential team member. Crit Care Med 2006;34:S46-51.
6. Kane SL, Weber RJ, Dasta JF. The impact of critical care pharmacists on enhancing patient outcomes. Intensive Care Med 2003;29:691-8.
7. Lat I, Paciullo C, Daley MJ, et al. Position paper on critical care pharmacy services: 2020 update. Crit Care Med 2020;48:e813-e834.
8. Leape LL, Cullen DJ, Clapp MD, et al. Pharmacist participation on physician rounds and adverse drug events in the intensive care unit. JAMA 1999;282:267-70.
9. Montazeri M, Cook DJ. Impact of a clinical pharmacist in a multidisciplinary intensive care unit. Crit Care Med 1994;22:1044-8.
10. Ovehage JM, Lukes A. Practical, reliable, comprehensive method for characterizing pharmacists' clinical activities. Am J Health Syst Pharm 1999;56:2444-50.
11. Patanwala AE, Narayan SW, Haas CE, et al. Proposed guidance on cost-avoidance studies in pharmacy practice. Am J Healty Syst Pharm 2021;78:1559-67.
12. Pawloski P, Cusick D, Amborn L. Development of clinical pharmacy productivity metrics. Am J Health Syst Pharm 2012;69:49-54.
13. Prelaski CR, Lat I, Poston J. Pharmacist contributions as members of the multidisciplinary ICU team. Chest 2013;144:1687-95.

ANSWERS AND EXPLANATIONS TO PATIENT CASES

1. **Answer: B**

Using evidence-based practice helps provide support in creating a clinical pathway with support and agreement among practitioners (Answer B is correct). Using closed-loop technology is important when implementing a clinical pathway, as is establishing a physician champion (Answer A). Developing a clinical protocol and pathway comes after the evidence is evaluated (Answer C). Formulary proposals are not a primary factor in implementing a clinical pathway (Answer D).

2. **Answer: C**

A gap analysis compares best practice and current state to identify opportunities for improvement (Answer C is correct). Creating a table of the ISMP best practice recommendations regarding storage and handling of NMBs and comparing them with current practice would be the best way to identify opportunity to improve. A survey could be helpful to learn what other institutions are doing, but it does not ensure they are in line with best practices. It could provide additional information but would not show where gaps exist in current state and best practice at your institution (Answer A is incorrect). Answer B is incorrect because it is jumping right to solutions without completing the gap analysis. Answer D is also incorrect because surveying the staff would not provide information on the best practice gaps that exist.

3. **Answer: B**

Answer A is incorrect because the patient name and dose do not need to be on the label per the NPSG requirements. Both drug name and expiration date/time should be. Answer B is correct because all of these must be on the label of a medication drawn up in the perioperative setting. Answer C is incorrect because the patient name and drug dose are not needed on the label but drug name and concentration do. Answer D is incorrect because drug dose does not need to be on the label.

4. **Answer: B**

Current data show that the critical care pharmacist affects many areas associated with TJC measures. According to TJC's current measures, the critical care pharmacist affects procoagulant reversal agent initiation for ICH (Answer B is correct). The ED median time from ED arrival to ED departure for admitted ED patients is not affected by the pharmacist (Answer A is incorrect). Hemorrhage transformation for patients treated with intravenous alteplase therapy or mechanical endovascular reperfusion therapy is also not affected by the pharmacist (Answer C is incorrect). Although the pharmacist would have impact on the severe sepsis and septic shock management bundle, this is not currently a required measurement (Answer D is incorrect).

5. **Answer: A**

Defining the problem is the first step in LEAN (Answer A is correct). The LEAN strategy involves attempting to remove the team members' opinions so their emotions do not influence the process. Hence, asking the nurses their opinion for the spike in catheter-related bloodstream infections influences their emotions (Answer B is incorrect). Identifying operational barriers is a LEAN process, however a transportation barrier is not related to catheter related blood stream infections (Answer C is incorrect). Describing the past process is not a LEAN evaluation component (Answer D is incorrect).

6. **Answer: C**

An MUE is drug- or disease-specific and is best suited for reviewing vancomycin dosing and the ordering of vancomycin blood concentrations (Answer C is correct). Quality assurance surveys are best suited for monitoring the medication use process that may not be specific to a drug or disease, such as the review of a pharmacist's notes in the EHR, review of the accuracy of expiration dates placed on intravenous admixture products, and review of the frequency of drug interaction warnings on the CPOE system (Answers A, B, and D are incorrect).

7. **Answer: A**

Justification to hospital leadership should provide financial benefit to the organization as well as improvement in patient outcomes and safety. Antibiotic stewardship outcomes as well as cost savings would provide both of these above and help justify additional pharmacy FTEs (Answer A is correct). Although provider education and in-services can be of benefit, it is difficult to justify these with cost savings or outcome benefits (Answer B is incorrect). Although many clinical

pharmacists spend time on rounds, this time is not value added without interventions and cost savings metrics associated with it (Answer c is incorrect). While renal dose adjustments and intravenous to oral conversion are interventions clinical pharmacists frequently make, they do not provide measurable metrics unless associated with a cost savings and safety benefit (Answer D is incorrect).

8. **Answer: C**

Answer A is incorrect because it solely focuses on cost with little rationale on how this might impact the patient clinically. Answer B is incorrect because it is worded such that it is implying harm will result if the provider does not take the recommendation. Answer C is correct because it is worded that the provider should consider the recommendation as well as the rationale as to why. Answer D is incorrect because it is focusing on the fact that it is non-formulary and would likely cause harm if the provider does not take the recommendation.

ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

1. **Answer: D**
Financial justification for a critical care pharmacist has been shown by the impact of clinical pharmacists on critically ill patients. The critical care pharmacist can reduce adverse events in the critically ill population, reduce order prescribing errors, optimize the correct drug for the correct disease process, decrease the LOS, reduce medication administration errors, and reduce inappropriate antibiotic use. One of the main financial impacts on critically ill patients is antimicrobial therapy. Using an anticoagulation reversal stewardship program will most affect the institution (Answer D is correct). Developing protocols is important in streamlining practice in the critically ill population and can reduce health care costs, and prescribing errors and medication administration errors can result in adverse events to the patient, but none of these would have the most significant effect on finances (Answers A, B, and C are incorrect).

2. **Answer: B**
Practice environment constraints include financial disincentives (not enough staff to support delirium screening), organizational constraints (screening process takes up valuable nursing time), perception of liability (if the screen is conducted and the test is positive, the provider will be blamed), and patient expectations (Answer B is correct). Another barrier to implementing delirium screening includes the key stakeholders' standard or routine work of the process, which is based on opinion, not evidence (Answer A is incorrect). A third barrier involves the professional context and includes clinical uncertainty. Practitioners may believe they have adequate skills to act. In addition, information overload occurs with the vast amount of literature being published. An example of a knowledge barrier includes providers being unfamiliar with how to perform delirium screening (Answer C is incorrect). A fourth barrier is when the key stakeholders' opinion does not agree with screening patients for delirium because of changes in evidence-based medicine and possible knowledge gaps (Answer D is incorrect).

3. **Answer: A**
Evaluating the use of pharmacotherapy in SUP in the ICU is best suited for an MUE (Answer A is correct). The goal of an MUE is to ensure optimal medication therapy management and improve patient safety and outcomes for drug-related processes—in this case, pharmacotherapy in SUP. Although one component of an MUE is PI, a review of the quality should occur before determining whether a PI project is necessary (Answer B is incorrect). An interventional MUE incorporates a review of quality in the form of making a pharmacotherapeutic intervention—PI. Although reviewing ADE data and medication error data from the ICU is helpful in detecting and determining problems associated with the use of pharmacotherapy in SUP, these are isolated events and are reporter-dependent, and a lack of reports does not ensure that the use of pharmacotherapy in SUP is appropriate (Answers C and D are incorrect). Only an MUE is a robust and comprehensive method of evaluating the use of pharmacotherapy in SUP.

4. **Answer: B**
Evaluating the management of warfarin-induced hypoprothrombinemia is best suited for an MUE. The goal of an MUE is to ensure optimal medication therapy management and improve patient safety and outcomes for drug-related processes; an MUE is drug-, drug class–, or disease-specific—in this case, management of warfarin-induced hypoprothrombinemia (Answer B is correct). Quality assurance is a process for monitoring the effectiveness and safety of the medication use process that includes prescribing, dispensing, and administering medications. Evaluating pharmacist verification times for routine orders in the ICU, drug interaction warnings on the CPOE system, and duplicate warnings on the CPOE system is not necessarily drug- or disease state–specific and is best suited for a QA review (Answers A, C, and D are incorrect).

5. **Answer: A**
The next best step is determining which patients with cardiac arrest should receive targeted temperature management. (Answer A is correct). Best practices have already been identified, so additional literature evaluation is not needed. (Answer B is incorrect). Finally, data will be collected on the current process of managing targeted temperature management (Answer D is incorrect), which will be evaluated to determine the areas for improvement in the outcome desired. The current process has already been determined, as provided in the question (Answer C is incorrect).

6. **Answer: D**
Because the data have already been collected, they must be analyzed to determine the baseline time to antibiotics (Answer D is correct). The measuring phase includes collecting the data, which has already been completed (Answer A is incorrect). Establishing methods is part of the design (Answer B is incorrect). The change process being implemented is the improvement phase (Answer C is incorrect).

7. **Answer: A**
Using weighted metrics for each critical care pharmacist activity – quantified over time and annualized – more accurately quantifies a critical care pharmacist's activities (Answer A is correct). The number of cardiac arrest codes attended each month, number and severity of hospital-acquired *Clostridioides difficile* infections each year, and the number of in-services provided each year are important, but do not quantify a comprehensive list of a critical care pharmacist's activities individually (Answers B, C, and D are incorrect).

8. **Answer: A**
Sepsis management bundle and VTE prophylaxis are quality measures reported to CMS (Answer A is correct). Thrombolytic therapy for patients with stroke is correct, but stress ulcer prophylaxis is not reported (Answer B is incorrect). Heart failure readmissions are reported, but pneumonia vaccination rates are not (Answer C is incorrect). Use of ventilator bundles is not reported, but rates of acute coronary syndrome readmission are reported (Answer D is incorrect).

Appendix 1. Centers for Disease Control and Prevention (CDC) – The Core Elements of Hospital Antibiotic Stewardship Programs: Antibiotic Stewardship Program Assessment Tool (excerpt)

CORE ELEMENTS OF HOSPITAL ANTIBIOTIC STEWARDSHIP PROGRAMS: ASSESSMENT TOOL		ESTABLISHED AT FACILITY	COMMENTS
Hospital Leadership Commitment	7. Does facility leadership support enrollment and reporting into the National Healthcare Safety Network (NHSN) Antimicrobial Use and Resistance (AUR) Module, including any necessary IT support?	❑ Yes ❑ No	
	8. Other example(s):	❑ Yes ❑ No	
Accountability	1. Does your facility have a leader or co-leaders responsible for program management and outcomes of stewardship activities?	❑ Yes ❑ No	
	a. If a non-physician is the leader of the program, does the facility have a designated physician who can serve as a point of contact and support for the non-physician leader?	❑ Yes ❑ No ❑ NA	
	2. Other example(s):	❑ Yes ❑ No	
Pharmacy Expertise	1. Does your facility have a pharmacist(s) responsible for leading implementation efforts to improve antibiotic use?	❑ Yes ❑ No	
	2. Does your pharmacist(s) leading implementation efforts have specific training and/or experience in antibiotic stewardship?	❑ Yes ❑ No	
	3. Other example(s):	❑ Yes ❑ No	

Exert from Source: Centers for Disease Control and Prevention (CDC). The Core Elements of Hospital Antibiotic Stewardship Programs: Antibiotic Stewardship Program Assessment Tool. Reference to specific commercial products, manufacturers, companies, or trademarks does not constitute its endorsement or recommendation by the U.S. Government, Department of Health and Human Services, or Centers for Disease Control and Prevention. Available at the following website at no charge. https://www.cdc.gov/antibiotic-use/healthcare/pdfs/assessment-tool-P.pdf.

Appendix 2. Example of Interventions and Clinical Pharmacy Services Cost Savings

Pharmacy Department
Data

Measure	Benchmark	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
Pharmacotherapeutic interventions	None	3451	5748	5015	7298
Outcomes cost and raw drug cost savings from pharmacotherapeutic interventions	None	$477,867	$681,967	$652,145	$701,542

Findings/Conclusion: For 2020, the total number of pharmacotherapeutic interventions remained the same from the previous year with 20,940 interventions, realizing $846,245 in outcomes and raw drug cost savings. For 2021, the total number of pharmacotherapeutic interventions remained similar to the prior year with 24,500 interventions, realizing $950,471 in outcomes and medication use savings. For 2022, the total number of pharmacotherapeutic interventions remained similar to the prior year with 23,452 interventions, realizing $904,254 in outcomes and raw drug cost savings. For 2023, the total number of pharmacotherapeutic interventions increased with 35,459 interventions documented, realizing $1,954,654 in outcomes and raw drug cost savings.

Pharmacotherapeutic interventions performed by clinical pharmacists consist of making downward and upward dosing adjustments of medications, providing pharmacokinetic consultations, avoiding drug-drug and drug-food interactions, avoiding toxic medications, avoiding drug-disease contraindications, avoiding drug-allergy interactions, approving and dosing restricted antibiotics, switching patients from intravenous medications to oral medications, initiating more effective or safer drug therapies, initiating equally efficacious but less expensive medications, discontinuing unnecessary and duplicate medications, changing dosage forms according to patient tolerance, switching nonformulary to formulary medications, and making recommendations to monitor for efficacy and toxicity.

Clinical pharmacists review and respond to abnormal drug blood concentration assays and laboratory values such as serum chemistry and coagulation profiles as they pertain to medication management. Notes documenting the interventions are placed in the pharmacy profile. Depending on the quantity of pharmacotherapeutic interventions and the resulting cost savings, the Pharmacy Department's efforts to document clinical interventions, ensure medication safety, and contain medication-related costs have been very effective.

Action Indicated: No additional actions are required. The Pharmacy Department will continue to perform and document clinical interventions and evaluate the resulting cost savings.